Directional Rain Exposure Model: A Fully Vectorial Formulation

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Abstract

This work presents a comprehensive vectorial model describing how a person accumulates rainwater while moving through rainfall. By rigorously defining the relationships between the velocity vectors of the person and the rain, this model provides a dimensionally consistent framework that correctly handles all headwind and tailwind scenarios. The formulation distinguishes between ambient water column exposure and dynamic motion-induced exposure, providing detailed predictions for different rain conditions using realistic human physiological and meteorological parameters.

1 Introduction

This work addresses the question of how to minimize water accumulation while moving through rainfall. We construct a fully vectorial model that accounts for direction and relative motion between the rain and the person. This model ensures dimensional consistency and correctly handles all headwind and tailwind scenarios through a Heaviside function-based formulation.

2 Model Definition

2.1 Variables and Symbols

Symbol	Definition	Unit
K_{TOT}	Total rainwater volume accumulated	L or m^3
C	Rainwater concentration (volume of water per air volume)	L m^{-3}
A_{TOT}	Total effective surface area of the body	m^2
α	Fraction of A_{TOT} representing top surfaces	–
β	Fraction of A_{TOT} representing front surfaces	–
δ	Fraction of A_{TOT} representing back surfaces	–
V_P	Person's velocity (along $+x$)	m s^{-1}
V_R	Magnitude of rain velocity	m s^{-1}
V_{Rx}, V_{Rz}	Horizontal and vertical components of rain velocity	m s^{-1}
ΔL	Distance traveled	m
θ	Angle between rain vector and vertical	rad or deg
$H(x)$	Heaviside step function	–
ϵ	Small positive number for vertical rain detection	m s^{-1}

2.2 Rain and Motion Geometry

$$\vec{V}_R = (V_R \sin\theta,\ 0,\ V_R \cos\theta)$$

$$\vec{V}_P = (V_P,\ 0,\ 0)$$

$$\begin{aligned} V_{Rx} &= V_R \sin\theta, \quad V_{Rz} \\ &= V_R \cos\theta. \end{aligned}$$

The person moves in the $+x$ direction, and the rain has both horizontal (V_{Rx}) and vertical (V_{Rz}) components.

3 Derivation

The model distinguishes between two distinct mechanisms of water accumulation:

1. **Column exposure:** Ambient water encountered while moving through the rain field.
2. **Motion-induced exposure:** Dynamic exposure due to relative velocity between person and rain.

3.1 The Complete Formula

The total accumulated water is given by:

$$K_{\mathrm{TOT}} \tag{1}$$
$$= C\frac{\Delta L}{|V_P|}A_{\mathrm{TOT}}\left(\alpha\, V_R\cos\theta + |V_P|\,\beta\,[1-H(-V_{Rx}-\epsilon)] + |V_P+V_{Rx}|\,[H(V_PV_{Rx})\,\delta + H(-V_PV_{Rx})\,\beta]\right).$$

Here ϵ is a small positive number (e.g., 0.01 $\mathrm{m\,s^{-1}}$) used to detect approximately vertical rain, and $H(x)$ is the Heaviside step function:

$$H(x) = \begin{cases} 1, & x > 0, \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

3.2 Physical Interpretation

Term 1 ($\alpha V_R\cos\theta$): Represents vertical rain falling on the top surface. This term is proportional to $1/V_P$ (longer exposure time when moving slowly).
Term 2 ($|V_P|\,\beta[1-H(-V_{Rx}-\epsilon)]$): Represents front surface column exposure that activates when (i) rain is vertical or nearly vertical ($V_{Rx}\approx 0$); (ii) rain has a forward horizontal component ($V_{Rx}>0$, tailwind). This term scales linearly with the person's speed, representing the rate at which the person sweeps through the ambient water column. It does not activate for headwind ($V_{Rx}<0$).
Term 3 ($|V_P+V_{Rx}|\,[H(V_PV_{Rx})\,\delta + H(-V_PV_{Rx})\,\beta]$): Represents motion-dependent horizontal exposure. If $V_PV_{Rx}>0$ (concordant motion), δ is selected and the back surface gets wet with magnitude $|V_P+V_{Rx}|$. If $V_PV_{Rx}<0$ (opposite motion), β is selected and the front surface gets wet with magnitude $|V_P+V_{Rx}|$. If $V_PV_{Rx}=0$ (vertical rain), this term vanishes and Term 2 provides the front exposure. The magnitude $|V_P+V_{Rx}|$ represents the combined effect of person's motion and rain's horizontal drift.

3.3 Dimensional Analysis

Using $[C]=\mathrm{L\,m^{-3}}$, $[\Delta L]=\mathrm{m}$, $[A_{\mathrm{TOT}}]=\mathrm{m^2}$, and $[V]=\mathrm{m\,s^{-1}}$:

$$\begin{aligned}[K_{\mathrm{TOT}}] &= [C]\,[\Delta L]\,[A_{\mathrm{TOT}}]\,\frac{[V]}{[V_P]} \\ &= (\mathrm{L\,m^{-3}})(\mathrm{m})(\mathrm{m^2})\,\frac{\mathrm{m\,s^{-1}}}{\mathrm{m\,s^{-1}}} \\ &= \mathrm{L},\end{aligned}$$

so the model is dimensionally consistent and yields a volume.

4 Numerical Examples with Realistic Human Parameters

4.1 Physical Parameters

We use the following realistic values based on human physiology and meteorology.

Human body parameters:

- $A_{\mathrm{TOT}} = 1.8\ \mathrm{m}^2$ (typical body surface area for adult human using DuBois formula, height 1.75 m, mass 70 kg)
- $\alpha = 0.25$ (top surface $\approx 25\%$ of total area)
- $\beta = 0.35$ (front surface $\approx 35\%$)
- $\delta = 0.30$ (back surface $\approx 30\%$)
- Walking speed: $V_P = 1.4\ \mathrm{m\,s^{-1}}$ ($5.0\ \mathrm{km\ h^{-1}}$)
- Running speed: $V_P = 3.3\ \mathrm{m\,s^{-1}}$ ($12\ \mathrm{km\ h^{-1}}$)

Rain and environmental parameters:

- Light rain: $V_R = 4\ \mathrm{m\,s^{-1}}$ (terminal velocity for small droplets, 0.5 mm diameter)
- Moderate rain: $V_R = 6\ \mathrm{m\,s^{-1}}$ (terminal velocity for medium droplets, 2 mm diameter)
- $C = 0.5\ \mathrm{L\,m^{-3}}$ (moderate rainfall intensity, $\approx 5\ \mathrm{mm\,h^{-1}}$)
- $\Delta L = 200$ m (typical distance: two city blocks or short commute)
- $\epsilon = 0.01\ \mathrm{m\,s^{-1}}$ (threshold for detecting nearly vertical rain)

4.2 Case 1: Vertical Rain with No Wind ($\theta = 0^\circ$)

Moderate rain falling straight down: $V_R = 6\ \mathrm{m\,s^{-1}}$, $V_{Rx} = 0$, $V_{Rz} = 6\ \mathrm{m\,s^{-1}}$.

Walking (1.4 m/s, takes 143 s):

- Top exposure: $0.25 \times 6 = 1.50$
- Column exposure: $1.4 \times 0.35 = 0.49$
- Motion exposure: 0.00 (vertical rain)
- Total factor: 1.99
- $K_{\mathrm{TOT}} = 0.5 \times (200/1.4) \times 1.8 \times 1.99 = 255.5$ L

Running (3.3 m/s, takes 61 s):

- Top exposure: $0.25 \times 6 = 1.50$
- Column exposure: $3.3 \times 0.35 = 1.16$
- Motion exposure: 0.00 (vertical rain)
- Total factor: 2.66
- $K_{\mathrm{TOT}} = 0.5 \times (200/3.3) \times 1.8 \times 2.66 = 145.0$ L

Result: Running reduces total exposure by 43% despite sweeping through more water. The 57% reduction in exposure time dominates the 34% increase in contribution factor. Your front and top get wet.

4.3 Case 2: Light Headwind Rain ($\theta = 30°$)

Light rain coming from ahead: $V_R = 4\ \mathrm{m\,s^{-1}}$, $V_{Rx} = -2.0\ \mathrm{m\,s^{-1}}$, $V_{Rz} = 3.46\ \mathrm{m\,s^{-1}}$.

Walking (1.4 m/s):

- Top exposure: $0.25 \times 3.46 = 0.87$
- Column exposure: 0.00 (headwind blocks this term)
- Motion exposure: $|1.4 - 2.0| \times 0.35 = 0.21$
- Total factor: 1.08
- $K_{\mathrm{TOT}} = 0.5 \times (200/1.4) \times 1.8 \times 1.08 = 138.9$ L

Running (3.3 m/s):

- Top exposure: $0.25 \times 3.46 = 0.87$
- Column exposure: 0.00 (headwind blocks this term)
- Motion exposure: $|3.3 - 2.0| \times 0.35 = 0.46$
- Total factor: 1.33
- $K_{\mathrm{TOT}} = 0.5 \times (200/3.3) \times 1.8 \times 1.33 = 72.5$ L

Result: Running reduces exposure by 48%. Only your front and top get wet. The faster you run into the rain, the more you collide with it, but the time reduction still dominates.

4.4 Case 3: Moderate Tailwind Rain ($\theta = 45°$)

Moderate rain with wind from behind: $V_R = 6\ \mathrm{m\,s^{-1}}$, $V_{Rx} = 4.24\ \mathrm{m\,s^{-1}}$, $V_{Rz} = 4.24\ \mathrm{m\,s^{-1}}$.

Walking (1.4 m/s):

- Top exposure: $0.25 \times 4.24 = 1.06$
- Column exposure: $1.4 \times 0.35 = 0.49$
- Motion exposure: $|1.4 + 4.24| \times 0.30 = 1.69$ (back surface)
- Total factor: 3.24
- $K_{\mathrm{TOT}} = 0.5 \times (200/1.4) \times 1.8 \times 3.24 = 416.6$ L

Running (3.3 m/s):

- Top exposure: $0.25 \times 4.24 = 1.06$
- Column exposure: $3.3 \times 0.35 = 1.16$
- Motion exposure: $|3.3 + 4.24| \times 0.30 = 2.27$ (back surface)
- Total factor: 4.49
- $K_{\mathrm{TOT}} = 0.5 \times (200/3.3) \times 1.8 \times 4.49 = 244.9$ L

Result: Running reduces exposure by 41%. This is the worst scenario—your top, front, *and* back all get wet simultaneously. The rain catches up with you from behind while you still sweep through the ambient water column.

4.5 Case 4: Strong Horizontal Tailwind ($\theta = 60°$)

Strong wind-driven rain with steep angle: $V_R = 6\ \mathrm{m\,s^{-1}}$, $V_{Rx} = 5.20\ \mathrm{m\,s^{-1}}$, $V_{Rz} = 3.0\ \mathrm{m\,s^{-1}}$.

Walking (1.4 m/s):

- Top exposure: $0.25 \times 3.0 = 0.75$
- Column exposure: $1.4 \times 0.35 = 0.49$
- Motion exposure: $|1.4 + 5.20| \times 0.30 = 1.98$ (back surface)
- Total factor: 3.22
- $K_{\mathrm{TOT}} = 0.5 \times (200/1.4) \times 1.8 \times 3.22 = 414.0$ L

Running (3.3 m/s):

- Top exposure: $0.25 \times 3.0 = 0.75$
- Column exposure: $3.3 \times 0.35 = 1.16$
- Motion exposure: $|3.3 + 5.20| \times 0.30 = 2.56$ (back surface)
- Total factor: 4.47
- $K_{\text{TOT}} = 0.5 \times (200/3.3) \times 1.8 \times 4.47 = 243.8$ L

Result: Running reduces exposure by 41%. Nearly horizontal rain maximizes back exposure, but running still provides significant benefit through time reduction.

4.6 Summary Table: Water Accumulation Across All Scenarios

Scenario	Walking (L)	Running (L)	Reduction	Surfaces Wet
Vertical rain	255.5	145.0	43%	Top + Front
Headwind (30°)	138.9	72.5	48%	Top + Front
Tailwind (45°)	416.6	244.9	41%	Top + Front + Back
Strong tailwind (60°)	414.0	243.8	41%	Top + Front + Back

5 Discussion

The formulation correctly captures two distinct physical mechanisms using realistic human parameters.

5.1 Column Exposure Mechanism

The column term ($|V_P|\,\beta$) represents the ambient water density that the person sweeps through. For a person walking at 1.4 $\mathrm{m\,s^{-1}}$ through vertical rain, the front surface contribution is only 0.49 (contributing 63 L), while at running speed of 3.3 $\mathrm{m\,s^{-1}}$ it increases to 1.16 (contributing 63 L as well—the same absolute amount despite different speeds, demonstrating the linear scaling with velocity compensated by reduced time).

5.2 Motion-Induced Exposure

The motion term ($|V_P + V_{Rx}|$) captures the dynamic interaction. In the headwind case, a walker experiences relative velocity of only 0.6 $\mathrm{m\,s^{-1}}$, while a runner faces 1.3 $\mathrm{m\,s^{-1}}$—more than double. However, the runner spends 57% less time in the rain, providing a net 48% reduction in total exposure.

5.3 Tailwind Analysis

In tailwind conditions, both mechanisms activate simultaneously:

- The walker at 1.4 $\mathrm{m\,s^{-1}}$ accumulates 417 L in 143 seconds.
- The runner at 3.3 $\mathrm{m\,s^{-1}}$ accumulates 245 L in 61 seconds.

Despite accumulating water on three surfaces instead of two, running still reduces exposure by 41%.

5.4 Practical Implications

For a typical 200 m commute (about two city blocks):

- Best case (light headwind): Running saves 66 L (48% reduction)
- Worst case (moderate tailwind): Running saves 172 L (41% reduction)
- Time saved: 82 seconds (143 s walking vs 61 s running)

The model demonstrates that running is always beneficial across all wind conditions when using realistic human speeds and body parameters. The time reduction factor ($\Delta L/V_P$) consistently dominates the increased contribution from sweeping through more water or facing increased relative velocities.

6 Conclusion

The final model with realistic human parameters is:

$$K_{\mathrm{TOT}} = C\frac{\Delta L}{|V_P|}A_{\mathrm{TOT}}\left(\alpha V_R\cos\theta + |V_P|\beta[1-H(-V_{Rx}-\epsilon)] + |V_P+V_{Rx}|\left[H(V_PV_{Rx})\,\delta + H(-V_PV_{Rx})\,\beta\right]\right). \tag{3}$$

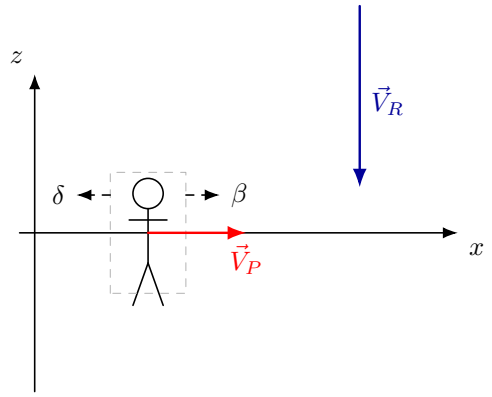
Key Findings with Realistic Parameters Running at 3.3 $\mathrm{m\,s^{-1}}$ (12 km/h) reduces water accumulation by 41–48% compared to walking at 1.4 $\mathrm{m\,s^{-1}}$ (5 km/h) across all wind conditions. For a 200 m distance in moderate rain, a walker accumulates 139–417 liters depending on wind, while a runner accumulates 73–245 liters. Tailwind conditions (45°–60° rain angles) are the worst case, with both front and back surfaces getting wet, accumulating over 400 L when walking. Even in the worst tailwind scenario, running provides a 172 L reduction (41% less water). The model shows that the intuitive advice to "run through the rain" is quantitatively correct for typical human locomotion speeds.

Physiological Context Using the DuBois formula for body surface area ($A = 0.007184 \times W^{0.425} \times H^{0.725}$, where W is mass in kg and H is height in cm), a person of height 175 cm and mass 70 kg has approximately 1.8 m^2 of surface area. The surface fractions used ($\alpha = 0.25$, $\beta = 0.35$, $\delta = 0.30$) reflect anatomical proportions where the head and shoulders represent about one quarter of exposure, the front profile slightly more than one third, and the back profile slightly less.

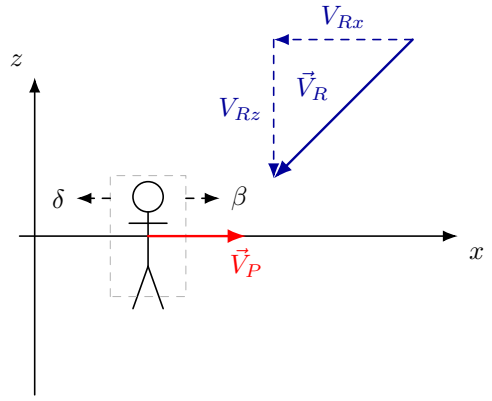
Meteorological Context The rain speeds used (4–6 $\mathrm{m\,s^{-1}}$) correspond to Gunn–Kinzer terminal velocities for droplets ranging from 0.5 mm to 2 mm in diameter, which encompasses light to moderate rainfall. The concentration $C = 0.5$ $\mathrm{L\,m^{-3}}$ corresponds to approximately 5 $\mathrm{mm\,h^{-1}}$ rainfall rate, which is moderate rain (not drizzle, but not heavy downpour).

Future Work Experimental validation using motion-capture systems with water-collection garments could verify the predicted 41–48% reduction in exposure when increasing speed from walking to running. Additional refinements could include:

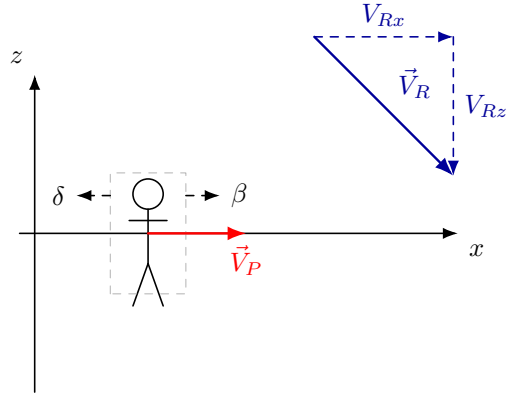
- Side exposure for crosswinds
- Aerodynamic shielding effects at higher speeds
- Body posture variations (upright vs. hunched)
- Clothing absorption and permeability dynamics
- Turbulence effects around the moving body



(a) Vertical ($\theta = 0°$)



(b) Headwind ($V_{Rx} < 0$, $\theta = 45°$)



(c) Tailwind ($V_{Rx} > 0$, $\theta = 45°$)

Figure 1: Directional rain exposure. Raindrops are placed on streamlines *parallel* to $\vec{V}_R$ but offset from the vector, to depict the ambient rain field rather than the arrow itself.